An Energy-Efficient Low-Voltage SRAM-based Charge Recovery Logic Near-Memory-Computing Macro for Edge Computing

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***Abstract*— In this paper, an energy-efficient low-voltage 7T SRAM-based charge recovery logic NMC macro is proposed. Firstly, a weight-stationary NMC macro architecture in dual clock and voltage domains is proposed to save memory energy consumption at near-threshold regime, without sacrificing computing throughput. Secondly, the charge recovery logic at subthreshold regime is also employed to reduce NMC logic energy consumption, while maintaining the computing speed. Simulation results show that the energy efficiency of the proposed 4Kb 7T-SRAM based CRL NMC macro design is around 3.71 TOPS/W, i.e., 6.49× improvement against the baseline design, when accelerating convolutional operations by 1.8 GOPS at 100 MHz with SRAM operating under 0.6 V and CRL computing under 0.4 V.**

Keywords—near memory computing, charge recovery logic, sub-threshold operation, near-threshold operation, SRAM

I. INTRODUCTION

In energy-efficient edge computing applications, digital computing-in-memory (CIM) architectures are normally divided into two major categories: In-Memory Computing (IMC) and Near-Memory Computing (NMC), as shown in Fig. 1. Recently, NMC has gained significant attention from researchers due to its ability of converting the dense data transfer of weights between storage arrays and IMC macros into a single storage array access [1], thereby significantly reducing data transfer overhead. Additionally, the NMC scheme supports more complex logic families and requires a simpler design effort, as compared to the IMC scheme that has a much tighter coupling between the storage and computing units.

In the literature, there are a few digital NMC designs operating based conventional logic such as dynamic logic in [1] and static logic in [2]. However, it is a significant challenge to design an ultra-low-power NMC macro, without sacrificing computing speed or accuracy for edge computing applications with stringent energy-constrained conditions. The NMC design in [3] utilizes stochastic computing to improve energy efficiency while sacrificing speed and accuracy. The NMC design in [4] employs approximate computing method to achieve high energy efficiency and maintain computing speed, but still trading off computing accuracy. Both the designs in [2] and [4] try to lower operating voltages to near-threshold voltages for reducing power consumption, while compromising of computing speed and stability. In summary, there are two main issues in the existing digital NMC designs:

1) Most existing designs adopt output-stationary (OS) data flow that partial sums must be continuously accumulated and stored in register buffers until a convolution operation is completed, which requires frequent weight read operations and results in intensive SRAM access power consumption up to 10-20% of total power in [4].

2) The existing designs utilize emerging logic families to achieve further energy efficiency improvement by trading off computing accuracy or speed, such as the stochastic logic and approximate logic based NMC designs in [3] and [4], respectively. Besides, aggressive voltage scaling to near-threshold or sub-threshold regimes can significantly achieve high energy efficiency, but the ultra-low-voltage NMC implementations inevitably suffers lower speed and poor stability.

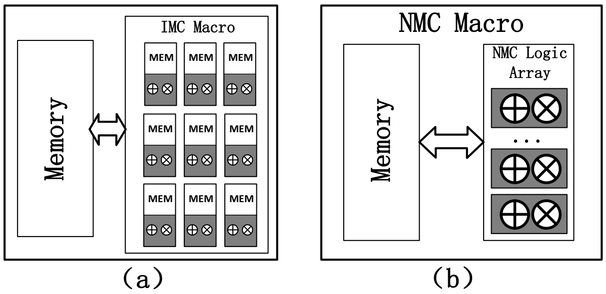

Fig. 1 Illustration of (a) in-memory computing and (b) near-memory computing architectures.

To solve above issues, we propose a low-voltage SRAM-based charge recovery logic NMC design scheme with two major features.

1) Firstly, a dual clock domain architecture based on weight-stationary (WS) data flow is proposed, which can reduce SRAM access frequency by leveraging weight reusability. Therefore, both the SRAM supply voltage and clock frequency can be significantly lowered for saving memory energy consumption. At the same time, the dual clock scheme enables the speed matching between the near-threshold SRAM in slow clock domain and the computing logic circuits in fast clock domain.

2) Secondly, the charge recovery logic family in [5] is employed in the NMC design, in which the computing power can be significantly reduced by operating the combination logic circuit at subthreshold voltage without sacrificing computing speed due to the latch-based pipeline design with voltage booster circuits in the charge recovery logic.

The remainder of this paper is organized as follows: Section II describes the architecture of CRL-based NMC macro. Section III presents implementation results,

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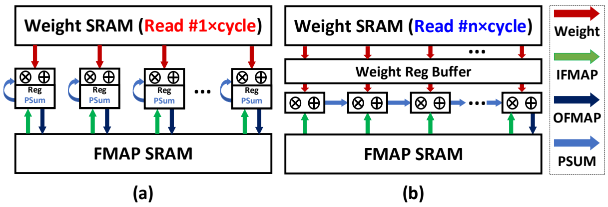

Fig. 2 Typical convolution computing architectures with (a) OS and (b) WS dataflows.

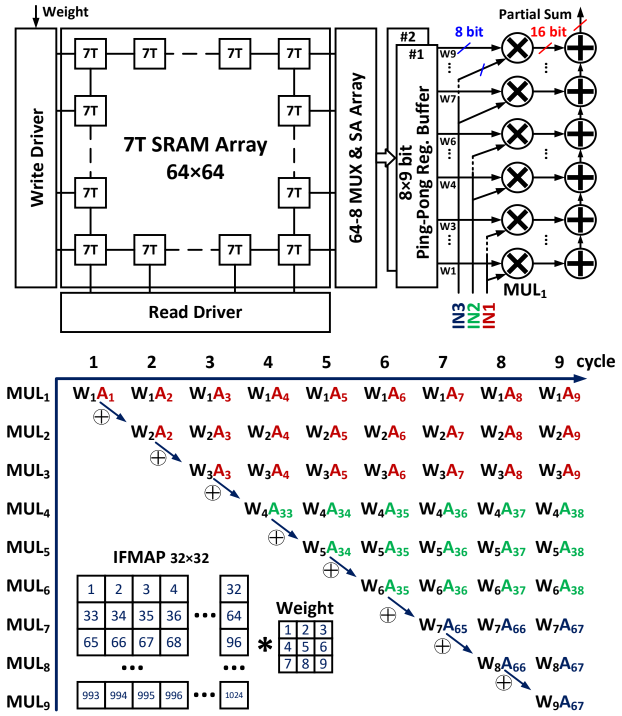

Fig. 3 (a) Architecture and (b) pipeline diagram of weight stationary pipeline NMC macro.

comparison and discussion. Section IV presents conclusion of this work.

II. Proposed Low-Voltage SRAM-Based Charge Recovery Logic NMC Macro

A. *Overall Weight Stationary Pipeline NMC Macro Architecture*

Fig. 2 (a) and (b) presents the two typical convolution computing architectures with OS and WS dataflows, respectively. The OS data flow requires reading different data of weights from weight buffer every cycle to calculate a pixel of output feature map (OFMAP) as soon as possible. In contrast, the WS data flow, which reuses weights data as much as possible, temporarily stores weight data in register buffers and updates new weight data from the weight buffer only after completing the MAC (Multiplication and ACcumulation) calculations associated with the stationary weights after multiple cycles. This feature allows the WS architecture to reduce SRAM access frequency to save energy consumption.

Fig. 3 (a) shows the proposed weight stationary Charge Recovery Logic (CRL) NMC macro architecture, consisting of a 7T near-threshold SRAM for storing weights of a convolutional layer (e.g., the 1st layer in LeNet model) in slow clock domain, a ping-pong register buffer for buffering weights of a 3×3 convolutional kernel channel, and a pipelined CRL array of 9 MAC units in fast clock domain.

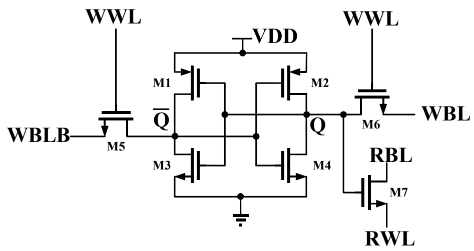

Fig. 4 7T SRAM cell structure which separates read and write operations and supports transposed readout.

The CRL array is implemented at two pipelining levels: i.e., inter-module pipelining between the multiplier and accumulator modules, and intra-module pipelining between the compressors/adders within the latch-based multiplier and accumulator modules. Therefore, this CRL array design can achieve a high pipeline clock frequency by implementing pipeline granularity down to the compressor level [6].

Fig.3 (b) shows the corresponding module-level pipeline diagram of the WS-based convolution computation. In general, after 9 weights of a convolution kernel are latched to the MAC arrays from the ping-pong buffer #1, the MAC calculations for three rows of data from one channel of the input feature map (IFMAP) are started in the fast clock domain, while the read of next 9 weights from the SRAM are started in the slow clock domain. Assuming the feature map width is W and computation clock cycle of the MAC unit is T_{MAC}, the ping-pong buffer #2 is allowed to read and latch next 9 weights to the MAC array after a time of W×T_{MAC}. Therefore, the weight SRAM access frequency can be reduced to $f_{SRAM} = f_{MAC}/W$ thanks to the unique property of WS data flow.

As shown in Fig. 3 (b), the partial sums obtained from MAC calculations shift right between the accumulators and are subsequently accumulated with the multiplication results from the next corresponding stage, finally producing the 3×3 convolution result at the ninth accumulator. When the 9 MAC units are fully pipelined, the ninth MAC unit can continuously output a convolution result every cycle in the fast clock domain. In this dual-clock scheme enabled by the WS data flow, the SRAM supply voltage is allowed to scale to near-threshold regime to save energy consumption, while the slow SRAM reading frequency can still match the the computing throughput of the MAC array.

Fig.4 presents the 7T SRAM cell structure. To ensure stability against the PVT variations at near-threshold regime, the 7T SRAM cell structure is employed to separate read and write operations to facilitate ultra-low-voltage SRAM design. Additionally, the 7T structure is also friendly to support transposed weight memory design, which can enable the transpose operations for on-chip CNN training [7-8].

B. *Charge Recovery Logic Multiplication and Accumulation Unit*

Fig. 5 (a) and (b) illustrate the basic structure of a CRL-based unit using a carry circuit of 3:2 compressor as an example, and the cascaded structure of CRL-based units, respectively. Each CRL-based unit consists of a sub-threshold all-NMOS complementary logic circuit for high-energy-efficiency logic computation and a latch-based boosting circuit for low-latency result transmission [5]. In the sub-threshold complementary logic circuit, NMOS devices are also used in the pull-up network (PUN) instead of PMOS to achieve gate-overdrive. For the latch-based

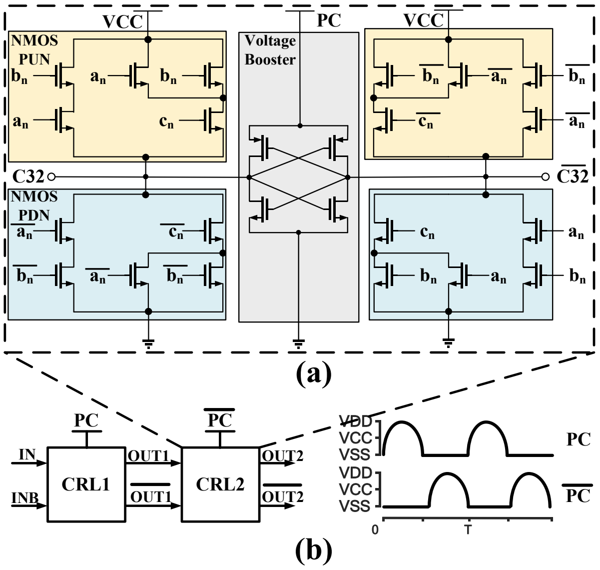

Fig. 5 (a) basic structure of a CRL-based unit and (b) cascaded structure of CRL-based units.

boosting circuit, the supply voltage is connected to a power clock (PC), which is generated by a blip power-clock generator to recover charge as the power clock falls, enabling energy-efficient operation. When the PC rises to the nominal voltage, the computation result of the logic part is latched by a pair of cross-coupled inverters as the PC reaches its high level. When multiple CRL-based units are cascaded to implement more complex logic, adjacent CRL-based units are connected with alternating voltage phases, which are PC and $\overline{PC}$, respectively.

Fig. 6 (a) shows the structure of the CRL-based MAC module consisting of a CRL-based MAC unit and a CRL-based output unit, in which the MAC unit results (i.e., SO and CO) are calculated and subsequently converted into square wave digital signals. Fig. 6 (b) and (c) illustrate the proposed DNN-oriented radix-4 Booth multiplier structure used in the MAC units and its arithmetic principle. The key idea is to exploit non-negativity of DNN IFMAP data (i.e., X) for reducing the number of sign compensation bits (i.e., SC) and also to compress the SC data as early as possible for reducing the number of compressing stages. Compared to the

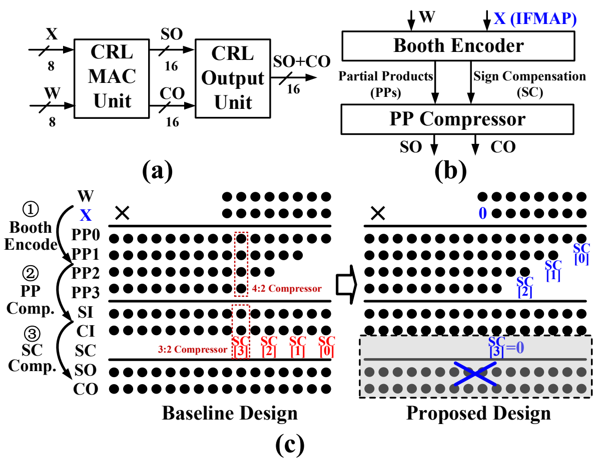

Fig. 6 Structure of (a) CRL-based MAC module and (b) proposed DNN-oriented radix-4 booth multiplier; (c) arithmetic principle of the proposed booth multiplier.

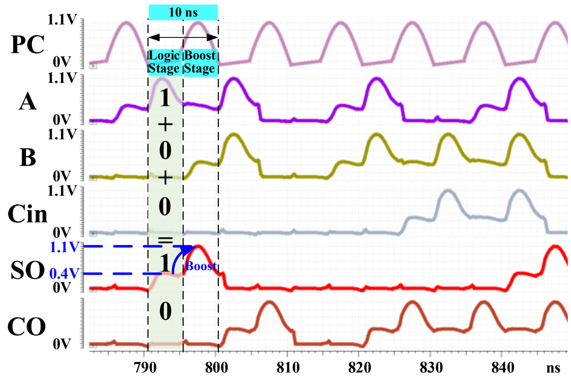

Fig. 7 Transistor-level simulation waveforms of CRL-based 3:2 compression.

baseline design in [6], the proposed design eliminates one compression stage, resulting in fewer compressors and CRL-based buffers. As shown in Fig. 6 (c), the baseline design uses a three-stage pipeline structure, which consists of a booth encoding stage encoding the multiplication inputs into 4 partial products (PPs) and 4 corresponding SC bits, a PP compression stage compressing the 4 PPs into SI and CI, and a SC compression stage compressing the SI, CI, and SC into SO and CO. In contrast to the baseline design, the proposed design not only leverages the non-negativity of the feature map data (i.e., X[7] = 0) to remove SC[3] (SC[3] = 0), but also compress the lower SC[2:0] with the 4 PPs in the PP compression stage, thereby eliminating the separate SC compression stage. In this way, the proposed design achieves a two-stage intra-module pipeline structure consisting of the basic building blocks such as half adder, 3:2 compressor and 4:2 compressor.

As a result, the proposed DNN-oriented radix-4 booth multiplier saves approximately 13 3:2 compressors and 6 buffer modules (i.e., approximately 592 transistors), as compared to the baseline design in [6].

III. Evaluation Results

To evaluate the energy efficiency of the proposed low-voltage SRAM-based CRL NMC macro, a 64×64 SRAM array and 9 CRL-based MAC units are designed and simulated as a case study with a 40-nm CMOS technology on Cadence Virtuoso. The SRAM operates at 10 MHz and under 0.6 V, and the CRL-based MAC units work at 100 MHz with 0.4 V for logic part and 1.1V for boosters. Besides, in order to demonstrate the energy saving with a fair comparison, a SRAM-based NMC macro baseline design using Synopsys DesignWare multiplier and adder IPs under the same WS data flow-based architecture is also implemented with both the SRAM and the MAC logic parts operate at 100 MHz under 1.1V.

Fig. 7 show the transistor-level simulation waveforms of 3:2 compression operation at 100 MHz under 0.4 V and 1.1 V, as an example to illustrate the CRL-based logic operation in logic and boost stages. It can be seen that when the PC signal is low (i.e., 0 V), the CRL logic circuit performs 3:2 compression computation at a sub-threshold voltage of 0.4 V in logic stage. As the PC signal rise to high (i.e., the nominal voltage 1.1 V), the CRL booster circuit latches the computation result at a rail-to-rail voltage at 1.1 V in boost stage.

Table I compares the energy efficiency between the proposed 7T SRAM-based CRL NMC macro and prior CIM designs. The stochastic computing-based design in [3]

TABLE I. COMPARISON WITH PRIOR CIM WORKS IN TERMS OF COMPUTING ENERGY EFFICIENCY

	ASSCC'22 [3]	ISSCC'22 [1]	Baseline Design	Proposed Design
Technology	65 nm	65 nm	40 nm	40 nm
Voltage(V)[a] SRAM/Logic	0.7-1.05/0.7-1.05	0.71-1.31/0.71-1.31	1.1/1.1	0.4&1.1/0.6
Frequency	3.2-5.2 MHz	25-125 MHz	100 MHz	100 MHz
Bit Precision [a]	A8W8	A3W3	A8W8	A8W8
Memory Cell Structure	10T SRAM	8T SRAM	7T SRAM	7T SRAM
Operation Mode	Digital In-memory	Digital Near-memory	Digital Near-memory	**Digital Near-memory**
Logic Type	Stochastic Logic	Dynamic Logic	Static Logic	**Charge Recovery Logic**
Memory Size	520 Kb	1152 Kb	4 Kb	4 Kb
Macro Throughput (GOPS)	N/A	200/0.74[b] @1.1 V, 100 MHz	1.8	1.8
Energy Efficiency (TOPS/W)	20 @0.7 V	0.829[c] @0.71 V	0.57	**3.71**

[a]A denotes Activation data and W denotes Weight data; [b]Scaled to a macro with 9 A8W8 MAC units [c]Scaled to A8W8

Fig. 8 (a) Power consumption breakdown and (b) power consumption comparison of the Synopsys DesignWare IP based NMC macro baseline design and the proposed low-voltage SRAM-based CRL NMC macro design.

achieves high energy efficiency at the cost of poor computation speed due to the binary to stochastic number conversion, which greatly limits the clock frequency (i.e., 3.2 MHz under 0.7 V). The dynamic logic based NMC macro design in [1] implements a 54-Kb SRAM, 432 multipliers and an adder tree, which achieves highest computational power. Our proposed design employing the WS dataflow and CRL logic can reduce the operating voltages of SRAM and logic cores to near/sub-threshold regimes without sacrificing computational power, and at the same time achieves higher energy efficiency.

Fig. 8 (a) and (b) show the power consumption breakdown and comparison of the NMC macro baseline design using Synopsys DesignWare IPs and the proposed design using low-voltage SRAM and CRL logic. Compared to the baseline design, the proposed design can achieve the same computational power but significantly reduces the SRAM read power consumption and NMC logic power consumption, by 76.7% and 84.9%, respectively. The energy efficiency improvement by 6.49× in terms of TOPS/W is attributed to the use of WS dataflow and the adoption of CRL logic in the proposed low-voltage NMC macro design.

IV. CONCLUSION

This paper proposes an energy-efficient low-voltage SRAM-based charge recovery logic NMC macro design. The NMC macro has two major design features: a weight-stationary NMC macro architecture with dual clock domains and near/sub-threshold supply voltages, and CRL-based NMC logic circuits operating in a latch-based pipeline fashion. Evaluation results demonstrate that the proposed design can achieve a significant energy efficiency improvement without sacrificing speed performance as compared to the baseline design.

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